

Multiplying and Dividing Decimals by Place-Value Reasoning

Place-value shifts, decimal products, decimal quotients, and estimating or comparing without computing

How to work:

- Say every number in place-value words first: 0.4 is *4 tenths*, 0.25 is *25 hundredths*.
- Multiply or divide the digits as if they were whole numbers, then decide where the decimal point belongs by thinking about the size of the answer.
- Show the place-value thinking, not just the algorithm. Label every answer with its unit.

1. One paper clip has a mass of 1.24 grams. A box holds 100 identical paper clips. Find the total mass of the paper clips in the box. Use the place-value chart to show how each digit moves.

hundreds	tens	ones	tenths	hundredths	thousandths
		1	2	4	

Answer: _____ g

2. A ribbon is 4.5 metres long. It is cut into 10 equal pieces. How long is each piece, in metres? Explain which way the digits shift and why the answer must be smaller than 4.5.

Answer: _____ m

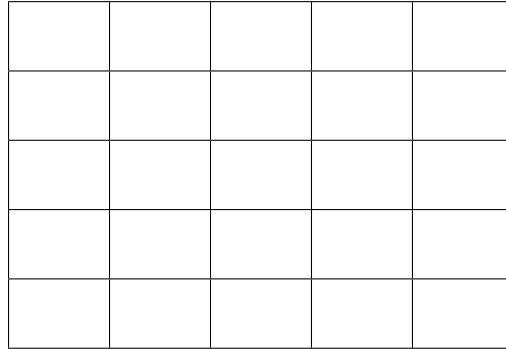
3. A soup recipe uses 0.4 litres of stock for one batch. The kitchen makes 6 batches. How many litres of stock are needed in all? Write the product first in tenths (for example, “4 tenths, six times”) and then as a decimal.

Answer: _____ L

4. A rectangular garden bed measures 6.8 metres by 4.2 metres. **Estimate** its area by rounding each measurement to the nearest whole metre, then multiplying. (Do not compute the exact area.) State the two rounded measurements you used.

Answer: _____ m²

- 5.** A wall tile measures 0.25 metres wide and 0.4 metres tall. Find its area in square metres. The area model below is one square metre cut into hundredths; use it to explain why the answer is a small fraction of a square metre.



Answer: _____ m²

- 6.** A rope 4.8 metres long is cut into pieces that are each 0.6 metres long. How many pieces are there? Show the place-value reasoning: rewrite both lengths in tenths, then divide.

Answer: _____

7. Without computing either value exactly, decide which is greater:

$$0.6 \times 12 \quad \text{or} \quad 12 \div 0.6$$

Write $<$ or $>$ between them, and explain your reasoning in one sentence using the words *smaller than one*.

Answer: _____

8. Nine identical books have a total mass of 7.2 kilograms. Each book has the same mass. Find the mass of one book, in kilograms. Show the division in tenths.

Answer: _____ kg

9. Find the mistake. A strip of fabric measures 1.4 metres by 0.35 metres. Sam multiplies and writes the area as 4.9 square metres.

- Use estimation to explain, in one sentence, why 4.9 cannot be correct.
- Give the correct area in square metres.

Answer: _____ m^2

10. A juice machine holds 8.4 litres of juice and fills cups that hold 0.35 litres each.

- (a) How many full cups can the machine fill from 8.4 litres?

Answer: _____

- (b) After 20 cups have been filled, how many litres of juice are left in the machine?

Answer: _____ L